

RSS-Based Access Point Localization Experiment

1 Experimental Setup

This experiment investigates the localization of a single Wi-Fi access point using received signal strength (RSS) measurements collected inside the Faculty of Engineering at Ferdowsi University of Mashhad. The target access point was located in a corridor area of the building. An independently measured coordinate was kept only as ground truth for final error evaluation:

36 deg 18'46.1"N, 59 deg 31'34.9"E

which is equivalent to:

latitude = 36.3128055556, longitude = 59.5263611111.

The building map was drawn from the available JSON map file. In particular, the room and corridor outlines were reconstructed from the polygon coordinates stored in the map data.

2 Data

The initial dataset used in this experiment is an Excel file containing only the measurements associated with the selected target access point:

selected_ap_b28ba99a8d2a_pivot_data.xlsx

This file is stored in the output directory of the experiment and contains two sheets:

- **selected_ap_samples**: the measurement records used for localization;
- **summary**: a compact summary of the selected access point and its RSS statistics.

The sample sheet contains the measurement coordinates, device and orientation metadata, and one RSS column for the selected access point. RSS measurements from other access points are not included. The RSS column is named:

selected_ap_rss_dbm

The selected access point is:

BSSID = b28ba99a8d2a

The valid measurements cover the corridor region around the selected access point and nearby spaces. Figure 1 shows the RSS samples overlaid on the building map. Stronger RSS values are shown using green shades, while weaker RSS values are shown using red shades.

Table 1: Summary of the Excel dataset used in the experiment.

Quantity	Value
Access point BSSID	b28ba99a8d2a
Number of valid RSS samples	1069
Maximum RSS	−29.0 dBm
Mean RSS	−72.89 dBm
Minimum RSS	−98.0 dBm

3 Initialization and Ground-Truth Evaluation

The ground-truth coordinate was not used to initialize or constrain the fitting procedure. As a simple RSS-only initialization, the measurement point with the maximum RSS was used as the initial AP position. For descriptive analysis, the RSS samples were also summarized by an RSS-weighted centroid. For a sample with RSS value r_i , the weight was defined as:

$$w_i = 10^{(r_i - r_{\max})/10}, \quad (1)$$

where $r_{\max}$ is the strongest RSS value in the dataset. The corresponding weighted centroid was computed as:

$$x_c = \frac{\sum_i w_i x_i}{\sum_i w_i}, \quad y_c = \frac{\sum_i w_i y_i}{\sum_i w_i}. \quad (2)$$

For this dataset, the RSS-weighted centroid was approximately 1.37 m away from the ground-truth coordinate. This indicates that the strongest RSS measurements are spatially concentrated around the expected access point location.

4 RSS Fitting Model

The access point location was estimated by fitting a log-distance path-loss model to the RSS measurements. The model is:

$$RSS(d) = RSS_{1m} - 10n \log_{10}(d), \quad (3)$$

where RSS_{1m} denotes the RSS at one meter, n is the path-loss exponent, and d is the Euclidean distance between a measurement point and the estimated access point position.

The unknown parameter vector was:

$$\theta = (x_{AP}, y_{AP}, RSS_{1m}, n). \quad (4)$$

For a measurement point i with local Cartesian coordinate (x_i, y_i) , the distance to a candidate AP location is:

$$d_i(\theta) = \max \left(\sqrt{(x_i - x_{AP})^2 + (y_i - y_{AP})^2}, 1 \right). \quad (5)$$

The lower bound of 1 m avoids numerical instability in the logarithm when a candidate AP location becomes very close to a measurement point. The model-predicted RSS at point i is then:

$$\hat{r}_i(\theta) = RSS_{1m} - 10n \log_{10}(d_i(\theta)), \quad (6)$$

and the residual is defined as:

$$e_i(\theta) = \hat{r}_i(\theta) - r_i, \quad (7)$$

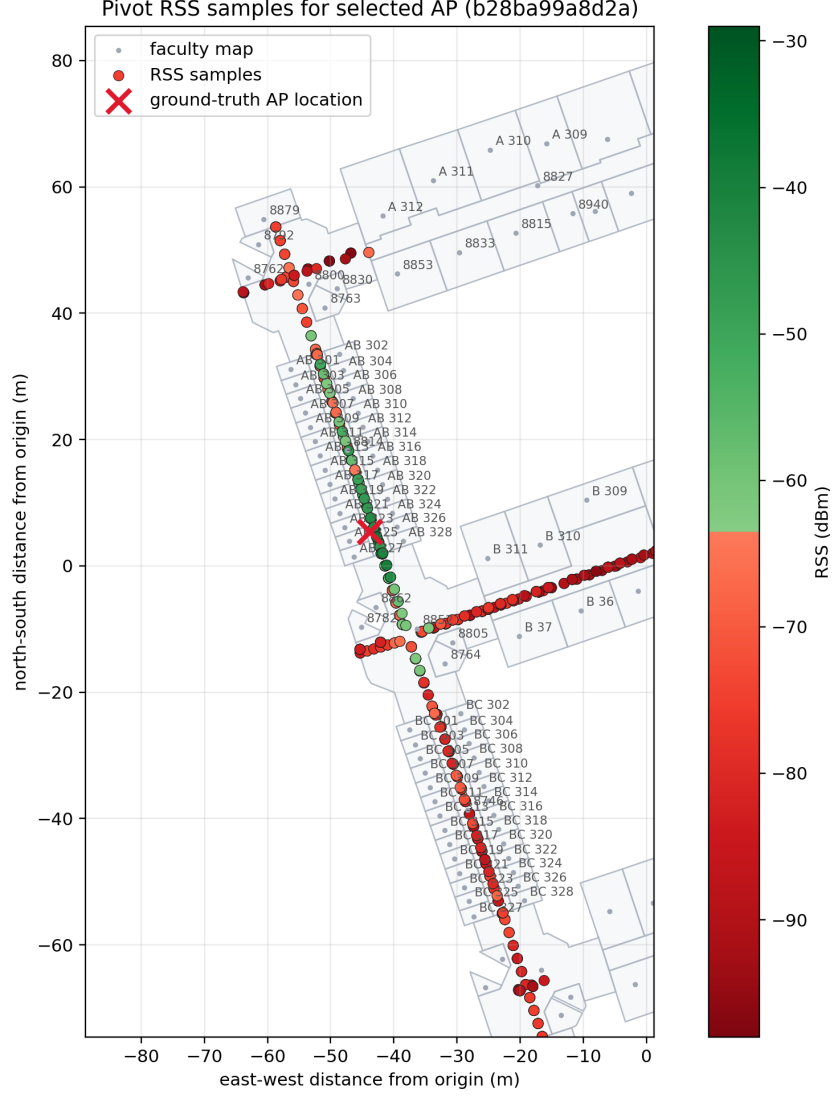


Figure 1: RSS measurements of the selected access point overlaid on the building map.

where r_i is the measured RSS value. Instead of minimizing the ordinary sum of squared residuals directly, a robust nonlinear least-squares objective was used:

$$\min_{\theta} \frac{1}{2} \sum_{i=1}^N \rho(e_i(\theta)^2). \quad (8)$$

The robust loss was the `soft_l1` loss, defined as:

$$\rho(z) = 2(\sqrt{1+z} - 1). \quad (9)$$

For small residuals, this loss behaves similarly to the squared loss. For large residuals, its growth is slower, so unusually noisy RSS samples have less influence on the final AP estimate. This is useful for indoor RSS measurements, where multipath propagation, shadowing, and device orientation can produce occasional outliers.

The fitting procedure was performed as follows:

1. First, all measurement coordinates were converted from latitude and longitude to a local Cartesian coordinate system in meters. This made it possible to compute Euclidean distances between the candidate AP location and each RSS measurement point.

2. Next, only valid RSS observations of the selected AP were used. In this dataset, valid observations are the 1069 samples listed in the Excel file.
3. The initial AP location for the optimizer was set to the measurement point with the maximum RSS value. This choice uses only the RSS dataset and does not use the ground-truth coordinate.
4. The initial value of RSS_{1m} was set from a high RSS percentile of the data, and the initial path-loss exponent was set to $n = 2$.
5. Bound constraints were used to keep the solution physically meaningful. The AP coordinates were allowed to move within the measurement region with an additional margin, RSS_{1m} was constrained to a realistic RSS interval, and the path-loss exponent was constrained to $0.5 \leq n \leq 6$.
6. For each candidate parameter vector $\theta = (x_{AP}, y_{AP}, RSS_{1m}, n)$, the distance from the candidate AP location to every measurement point was computed. The log-distance model then predicted the RSS value at each point.
7. At each iteration, the optimizer evaluated the residual vector $\mathbf{e}(\theta)$, applied the robust loss, and updated the four unknowns $(x_{AP}, y_{AP}, RSS_{1m}, n)$ to reduce the objective function. The process continued until the change in the objective became sufficiently small.
8. Finally, the optimized AP coordinate was converted back from the local Cartesian coordinate system to latitude and longitude. The ground-truth coordinate was used only at this final stage to compute the localization error.

5 Localization Results

Table 2 reports the fitted model parameters and the estimated access point location.

Table 2: Estimated AP location and fitted RSS model parameters.

Quantity	Value
Initial latitude	36.3127978000
Initial longitude	59.5263734000
Estimated latitude	36.3128106176
Estimated longitude	59.5263414105
Distance from ground truth	1.85 m
Estimated RSS_{1m}	-28.36 dBm
Path-loss exponent n	3.22
RSS RMSE	8.53 dB
RSS MAE	6.91 dB

Figure 2 shows the ground-truth coordinate and the estimated location on the building map. The red cross marks the ground-truth coordinate, and the black star marks the estimated access point position.

The estimated location was approximately 1.85 m away from the ground-truth coordinate. Given the variability of indoor RSS measurements, this result shows that the log-distance fitting model provides a practical approximation of the access point position when a sufficient number of measurements are available in the surrounding corridor area.

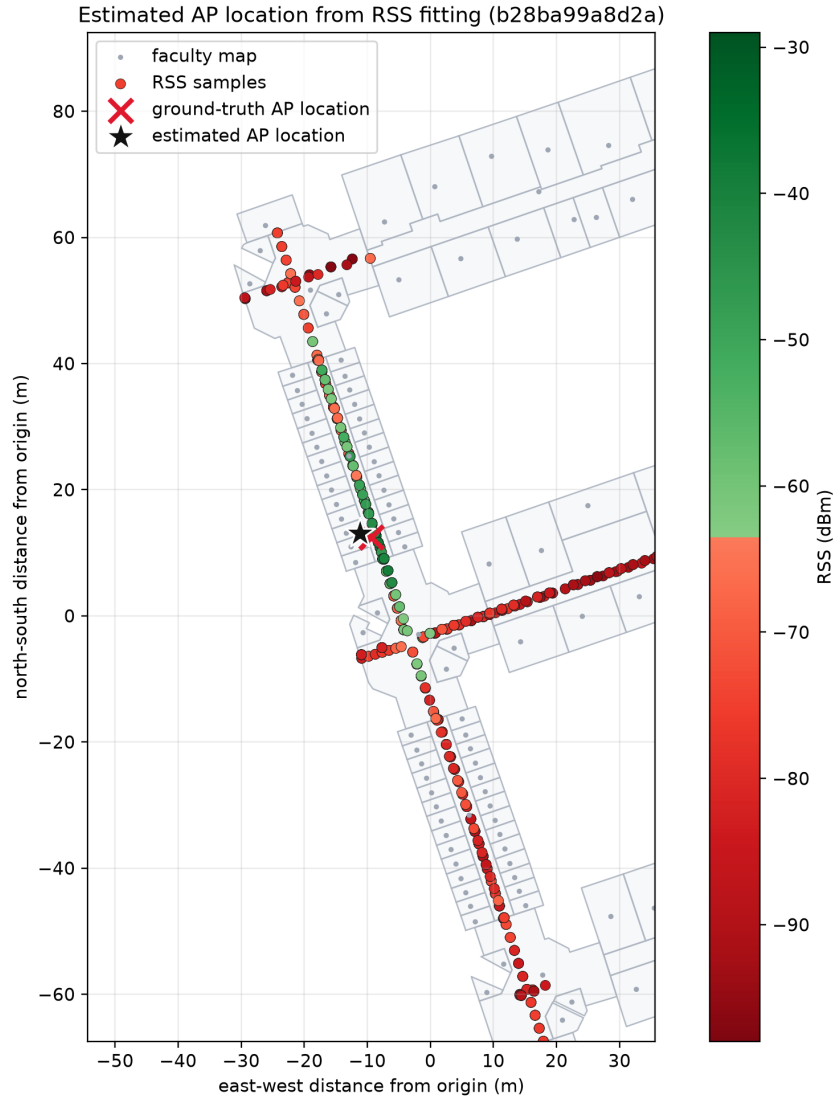


Figure 2: Reference and estimated access point locations on the building map.